

Question:	1	2	3	4	Total
Points:	6	8	8	8	30
Score:					

Instructor: Dr. Ali Alsaffar.

(1) Answer the following questions.

(a) [2 points] Give a name of an algorithm for each of the following problem types.

Problem Type	Algorithm
Sorting	<i>Selection Sort, Bubble Sort</i>
Combinatorial	<i>Traveling Salesman Problem (TSP), Graph Coloring</i>
Geometric	<i>Closest Pair Problem, Convex-hull Problem</i>
Numerical	<i>GCD, Computing definite integrals, Computing Square Root, etc.</i>

(b) [2 points] Why time in seconds or milliseconds is not suitable to measure the time efficiency of algorithms? List two reasons.

Solution:

1. Dependence on Computer speed.
2. Dependence on Quality of implementation (coding).
3. Dependence on the compiler generating the code.
4. Difficulty of clocking the actual running time.

(c) [2 points] For the *string matching* algorithm, give an example of a text T and a pattern P for the worst case and compute how many key comparisons are executed for your example.

Solution: Let $T = AAAAAA$ and $P = AAB$.

If n and m are the sizes of T and P , respectively, then the number of comparisons is $(n - m + 1)m$. For the above example, the algorithm makes $(6 - 3 + 1)(3) = 4 \times 3 = 12$ comparisons.

(2) [8 points] Write a linear algorithm $O(n)$ with a single loop to compute the value of $\ln(1 + x)$ from the series:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots + \frac{x^n}{n}$$

The algorithm should add n terms with alternating signs (positive and negative), to get the approximate value of $\ln(1+x)$.

Solution:

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1: Algorithm LN( $x, n$ )
2:   //Input: A real number  $x$  and a positive integer  $n$ 
3:   //Output: The value of  $\ln(1+x)$ 
4:    $sum \leftarrow x$ 
5:    $power \leftarrow x$ 
6:    $sign \leftarrow 1$ 
7:   for  $i \leftarrow 2$  to  $n$  do
8:      $sign \leftarrow -sign$ 
9:      $power \leftarrow power * x$ 
10:     $sum \leftarrow sum + sign * power / i$ 
11:  return  $sum$ .
```

(3) Consider the following program segment.

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1:  $i \leftarrow 3, p \leftarrow 1$ 
2: while  $i \leq 3n$  do
3:   if  $i > n + 1$  then
4:      $p \leftarrow p * i$ 
5:    $i \leftarrow i + 1$ 
```

(a) [4 points] Find number of times the comparison $C(n)$ at line 2 is executed? Use summations to express your answer.

Solution:

$$C(n) = 1 + \sum_{i=3}^{3n} 1 = 1 + 3n - 3 + 1 = 3n - 1$$

(b) [4 points] Find number of times the multiplication $M(n)$ at line 4 is executed? Use summations to express your answer.

Solution:

$$M(n) = \sum_{i=n+2}^{3n} 1 = 3n - (n + 2) + 1 = 2n - 1$$

(4) Consider the following recursive algorithm.

```

1: Algorithm F( $n$ )
2:   if ( $n = 1$ ) then
3:     return 1
4:   else
5:      $s_1 \leftarrow n * F(n/2)$ 
6:      $s_2 \leftarrow n + F(n/2)$ 
7:   return  $s_1 + s_2$ 
```

(a) [3 points] Write the recurrence relation for the time complexity of the algorithm $T(n)$.
Hint. What is the basic operation?

Solution: The basic operation is the multiplication at line 5.
The time complexity recurrence $T(n)$:

$$T(n) = \begin{cases} 0 & n = 1 \\ 0 + 2T(n/2) + 1 & n > 1 \end{cases}$$

- (b) [5 points] Find the solution of $T(n)$ and determine its growth rate.

Solution:

$$\begin{aligned} T(n) &= 2T(n/2) + 1 && \leftarrow \text{step 1} \\ &= 2^2T(n/2^2) + 2 + 1 && \leftarrow \text{step 2} \\ &= 2^3T(n/2^3) + 2^2 + 2 + 1 && \leftarrow \text{step 3} \\ &\vdots \\ &= 2^kT(n/2^k) + \sum_{j=0}^{k-1} 2^j && \leftarrow \text{step } k \\ &= 2^kT(n/2^k) + 2^k - 1 \end{aligned}$$

Until $T(n/2^k) = T(1)$, $n/2^k = 1 \implies n = 2^k \implies k = \log_2 n$, and
 $\therefore T(n) = nT(1) + n + 1 = n + 1 \in \Theta(n)$.